

Senna Na

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Senior robotics and motion systems engineer with 7+ years of experience developing nanometer-precision electromechanical systems.

Skills : System integration, Root cause analysis, Reliability/FMEA, Requirements development, Cross-functional technical leadership

Technologies : MATLAB, R, Python, Siemens NX, CATIA, GD&T, Polarion, DMM, Oscilloscope, SEM, Sensor calibration and testing

Work Experience

ASML, Wilton CT

Senior Mechatronics Engineer / Functional Architect

Oct 1, 2024 - Present

- Own the system architecture, requirements, reliability, and roadmap for a piezo-actuated nanometer-precision mechanism
- Direct root cause analysis, corrective actions, and customer communication for critical worldwide customer escalations (1 per week), leading a 10-person team across mechanical, controls, software, electrical, supply chain, and sourcing disciplines
- Interface with Tier 1 hardware suppliers and software farmouts by contributing to statement of work, and ongoing alignments
- Evaluate tradeoffs, identify risks and mitigation, accelerated lifetime testing, and lead technical design reviews and FMEAs
- Deliver clear presentations to executive program management and external customers, including via international travel

Mechatronics Engineer / Integration Lead, Safety, Escalations

Jan 13, 2020 - Sep 30, 2024

- Led system integration, bring-up, test plan creation, and deployment of a new 6-DOF motion control platform, coordinating 30+ cross-functional engineers with hands-on troubleshooting of magnetic actuators, optical encoders, differential and capacitive sensors, linear/non-linear calibration, firmware, software, and communication via DMM, oscilloscope, accelerometers, and SEM
- Designed a patented safety-critical electromagnetic protection system in firmware and software, enabling +20% throughput. Created a state-space model of the plant in SIMULINK and performed Monte Carlo analysis with experimental data correlation
- Developed fleet data analysis and visualization tools and scripting in Python, MATLAB, and R to provide data-driven insights
- Improved system performance and statistical process control (SPC) through control-loop optimization such as setpoint design, frequency-domain (FFT) and transfer function analysis, bandwidth, PID, and feedforward tuning, and filtering techniques

Mechatronics Engineering Intern

Aug - Dec 2018, June - Sep 2019

- Developed a finite-state machine to optimize multi-robot throughput to improve system efficiency under varying conditions
- Performed system identification and modal testing, including DAQ setup and fixture design in Siemens NX via Teamcenter PLM

Honda Aircraft Company, Greensboro NC

Hydraulics and Landing Gear Intern

May 29, 2018 - Aug 17, 2018

- Created a MATLAB thermal model to define the airplane turnaround constraints by analyzing flight test and supplier data
- Supported engineering by developing a vibration test plan per DO-160, fault isolation trees, and CATIA GD&T drawing updates

Education

- 2021 - 2024 **Columbia University, M.S. Mechanical Engineering, *Robotics and Controls*.** GPA : 3.9
Robot Learning (Python), Evolutionary Algorithms (C++, MuJoCo), Robotics Studio (ROS), Computer Vision II (Python)
- 2016 - 2019 **Cornell University, B.S. Mechanical Engineering.** GPA : 4.0, Summa cum laude
Autonomous Mobile Robots (Kalman filtering), Feedback Control Systems, Formal Methods for Robotics

Research

Research Assistant, Biorobotics and Locomotion Lab, Ithaca NY

Jan 24, 2018 - May 22, 2018

- Designed a knee stop for damage protection by iteratively simulating FEM to achieve a target deflection under a maximum load
- Manufactured composite-foam-and-rubber mechanical stops with 3D molds and robot feet via carbon fiber wet layup

Research Assistant, Zehnder Research Group, Ithaca NY

Jan 25, 2017 - May 23, 2017

- Analyzed microscope images and CT scans on ImageJ for manufacturing defects and ran tension tests on carbon-fiber tubes

Projects

Front Assembly Co-Lead, Cornell Electric Vehicles, Ithaca NY

Aug 21, 2017 - Dec 31, 2019

- Co-led a team of 5 to redesign (CAD) the steering system dynamics, manufacture (machine, CNC, 3D print), assemble, and test
- Designed a topologically optimized load support web (waterjet cutting) and a generatively-designed brake pedal (DMLS)
- Designed a steering wheel by researching driver UX and 3D-print threads; in v2, studied print orientation and fits for phone dash